

COMP9020 23T1

Week 5

Graphs

- Textbook (R & W) - Ch. 3, Sec. 3.2
Ch. 6, Sec. 6.1-6.5
- A. Aho & J. Ullman. *Foundations of Computer Science in C*,
p. 522–526 (Ch. 9, Sec. 9.10)
- Problem set 5 + quiz
- **Mid-term test on Friday Week 6**

Mid-term Test

1 hour (+5mins reading time) online test on Moodle
Friday, 24 March between 10:30am and 11:40am

⇒ **must** start no later than by 10:35am to get the full 1hr+5mins

Format:

- Same environment as weekly quizzes but **strictly timed**
- 7 questions, each worth between 2 and 7 marks
maximum marks: 26
- mix of multiple choice (with ≥ 1 correct answers),
numerical/short answer and *open answer*

NB

The test will be manually marked, unlike the quizzes.

Mid-term Test

If you

... are uncertain about how to interpret a question

... are unsure about how to answer a question

... find a question too difficult

⇒ do answer to the best of your understanding

⇒ do focus on the questions that you find easier

⇒ do not agonise about a question or your answer after you've submitted

Practice test with 3 sample questions available in Moodle

- 1hr time limit, opens Friday, closes **Thursday week at 10am**
- sample solutions to open questions will be provided

Graphs

Terminology (the most common; there are many variants):

(Undirected) Graph — pair $G = (V, E)$ where

V – set of vertices

E – set of edges

Every edge $e \in E$ corresponds uniquely to the set (an unordered pair) $\{x_e, y_e\}$ of vertices $x_e, y_e \in V$.

A *directed* edge is called an *arc*; it corresponds to the ordered pair (x_a, y_a) . A **directed graph** consist of vertices and arcs.

NB

Edges $\{x, y\}$ and arcs (x, y) with $x = y$ are called *loops*.

We will only consider graphs without loops.

NB. Binary relations on finite sets correspond to directed graphs. Symmetric relations correspond to undirected graphs.

Graphs in Computer Science

Examples

- 1 The WWW can be considered a massive graph where the nodes are web pages and arcs are hyperlinks.
- 2 The possible states of a program form a directed graph.
- 3 The map of the earth can be represented as an undirected graph where edges delineate countries.

NB

Applications of graphs in Computer Science are abundant, e.g.

- route planning in navigation systems, robotics
- optimisation, e.g. timetables, utilisation of network structures
- compilers using “graph colouring” to assign registers to program variables

Vertex Degrees

- **Degree** of a vertex v in a graph G

$$\deg(v) = |\{ w \in V(G) : (v, w) \in E(G) \}|$$

i.e., the number of edges attached to the vertex

- **Regular graph** — all degrees are equal
- *Degree sequence* $D_0, D_1, D_2, \dots, D_k$
where D_i = no. of vertices of degree i

Question

What is $D_0 + D_1 + \dots + D_k$?

- $\sum_{v \in V} \deg(v) = 2 \cdot |E(G)|$; thus the sum of vertex degrees is always even.
- There is an even number of vertices of odd degree (6.1.8)

Paths

- A **path** in a graph (V, E) is a sequence of edges that link up

$$v_0 \xrightarrow{\{v_0, v_1\}} v_1 \xrightarrow{\{v_1, v_2\}} \dots \xrightarrow{\{v_{n-1}, v_n\}} v_n$$

where $e_i = \{v_{i-1}, v_i\} \in E$

- **length** of the path is the number of edges: n
neither the vertices nor the edges have to be all different
- Subpath of length r : $(e_m, e_{m+1}, \dots, e_{m+r-1})$
- Path of length 0: single vertex v_0
- **Connected graph** — each pair of vertices joined by a path
- **Connected component** of G — a connected subgraph of G that is not contained in a larger connected subgraph of G

Examples

Exercise

6.1.13(a) Draw a connected, regular graph on four vertices, each of degree 2

6.1.13(b) Draw a connected, regular graph on four vertices, each of degree 3

6.1.13(c) Draw a connected, regular graph on five vertices, each of degree 3

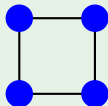
6.1.14(a) Graph with 3 vertices and 3 edges

6.1.14(b) Two graphs each with 4 vertices and 4 edges

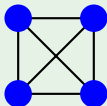
Examples

Exercise

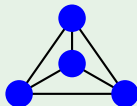
6.1.13 Connected, regular graphs on four vertices



(a)



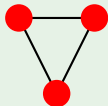
(b)



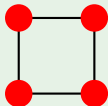
(b)

none
(c)

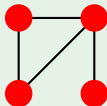
6.1.14 Graphs with 3 vertices and 3 edges must have a *cycle*



(a) the only one



(b)



(b)

Exercise

6.1.20(a) Graph with $|E(G)| = 21$ edges has a degree sequence $D_0 = 0, D_1 = 7, D_2 = 3, D_3 = 7, D_4 = ?$
Find $|V(G)|$!

6.1.20(b) How would your answer change, if at all, when $D_0 = 6$?

Exercise

6.1.20(a) Graph with $|E(G)| = 21$ edges has a degree sequence $D_0 = 0, D_1 = 7, D_2 = 3, D_3 = 7, D_4 = ?$

Find $|V(G)|$

$\sum_v \deg(v) = 2|E|$; here

$7 \cdot 1 + 3 \cdot 2 + 7 \cdot 3 + x \cdot 4 = 2 \cdot 21$ giving $x = 2$, thus

$|V| = \sum D_i = 19$.

6.1.20(b) How would your answer change, if at all, when $D_0 = 6$?
No change to D_4 ; $|V| = 25$.

Cycles

Recall paths $v_0 \xrightarrow{e_1} v_1 \xrightarrow{e_2} \dots \xrightarrow{e_n} v_n$

- *simple path* — $e_i \neq e_j$ for all edges of the path ($i \neq j$)
- *closed path* — $v_0 = v_n$
- **cycle** — closed simple path, all v_i distinct (except $v_0 = v_n$)
- *acyclic path* — $v_i \neq v_j$ for *all* vertices in the path ($i \neq j$)

NB

- ① $C_n = (e_1, \dots, e_n)$ is a cycle *iff* removing *any* single edge leaves an acyclic path. (Show that the ‘any’ condition is needed!)
- ② C is a cycle if it has the same number of edges and vertices and *no* proper subpath has this property.
(Show that the ‘subpath’ condition is needed, i.e., there are graphs G that are **not** cycles and $|E(G)| = |V(G)|$; every such G must contain a cycle!)

Trees

- **Acyclic graph** — graph that doesn't contain any cycle
- **Tree** — connected acyclic graph
- A graph is acyclic *iff* it is a *forest* (collection of unconnected trees)

NB

Graph G is a tree

- ⇔ G is acyclic and $|V(G)| = |E(G)| + 1$.
(Show how this implies that the graph is connected!)
- ⇔ there is exactly one simple path between any two vertices.
- ⇔ G is connected, but becomes disconnected if any single edge is removed.
- ⇔ G is acyclic, but has a cycle if any single edge on already existing vertices is added.

Exercise

6.7.3 (supp) Tree with n vertices, $n \geq 3$.

Always true, false or could be either?

(a) $|E(T)| \stackrel{?}{=} n$

(b) at least one vertex of deg 2

(c) at least two v_1, v_2 s.t. $\deg(v_1) = \deg(v_2)$

Exercise

6.7.3 (supp) Tree with n vertices, $n \geq 3$.

Always true, false or could be either?

- (a) $|E(T)| \stackrel{?}{=} n$ — False
- (b) at least one vertex of deg 2 — Could be either
- (c) at least two v_1, v_2 s.t. $\deg(v_1) = \deg(v_2)$ — True

NB

A tree with one vertex designated as *root* is called a **rooted tree**. It imposes an ordering on the edges: 'away' from the root — from parent nodes to children. This defines a *level number* of a node as its distance (= path length) from the root.

The *height* of a rooted tree is the largest level number of a vertex.

Another very common notion in Computer Science is that of a *DAG* — a *directed, acyclic graph*.

Graph Isomorphisms

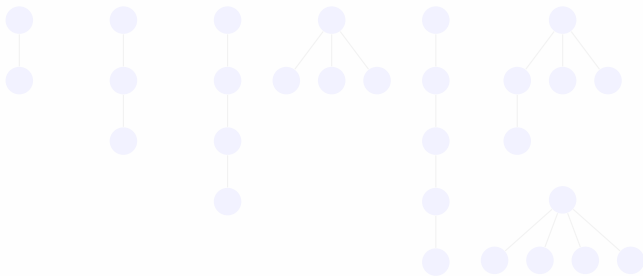
$\iota : G \longrightarrow H$ is a *graph isomorphism* if

- (i) $\iota : V(G) \longrightarrow V(H)$ is 1-1 and onto (a so-called *bijection*)
- (ii) $\{x, y\} \in E(G)$ iff $\{\iota(x), \iota(y)\} \in E(H)$

Two graphs are called *isomorphic* if there exists (at least one) isomorphism between them.

Example

All nonisomorphic trees on 2, 3, 4 and 5 vertices.



Graph Isomorphisms

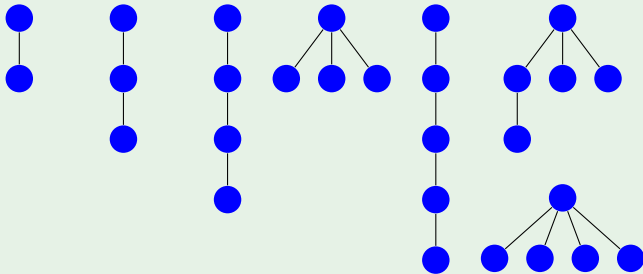
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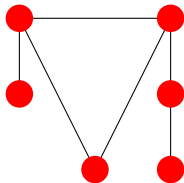
All nonisomorphic trees on 2, 3, 4 and 5 vertices.



Automorphisms and Asymmetric Graphs

An isomorphism from a graph to itself is called *automorphism*. Every graph has at least the trivial automorphism (trivial means: $\iota(v) = v$ for all $v \in V(G)$)

Graphs with no non-trivial automorphisms are called *asymmetric*. The smallest non-trivial asymmetric graphs have 6 vertices.



(Can you find another one with 6 nodes? There are seven more.)

Edge Traversal

Definition

- **Euler path** — path containing every edge of G exactly once
- **Euler circuit** — closed Euler path

(Named after Leonhard Euler (Switzerland), 1707–1783)

Characterisations

- G (connected) has an Euler circuit iff $\deg(v)$ is even for all $v \in V(G)$.
- G (connected) has an Euler path iff either it has an Euler circuit (above) or it has exactly two vertices of odd degree.

NB

- These characterisations apply to graphs with loops as well
- For directed graphs the condition for existence of an Euler circuit is $\text{indeg}(v) = \text{outdeg}(v)$ for all $v \in V(G)$

Special Graphs

- **Complete graph** K_n

n vertices, all pairwise connected, $\frac{n(n-1)}{2}$ edges.

- **Complete bipartite graph** $K_{m,n}$

Has $m + n$ vertices, partitioned into two (disjoint) sets, one of n , the other of m vertices.

All vertices from different parts are connected; vertices from the same part are disconnected. No. of edges is $m \cdot n$.

- **Complete k -partite graph** $K_{m_1, \dots, m_k}$

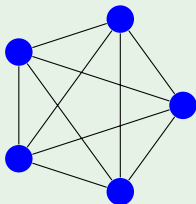
Has $m_1 + \dots + m_k$ vertices, partitioned into k disjoint sets, respectively of $m_1, m_2, \dots$ vertices. All vertices from different sets are connected; vertices from the same set are

disconnected. No. of edges is $\sum_{i < j} m_i m_j = \frac{1}{2} \sum_{i \neq j} m_i m_j$

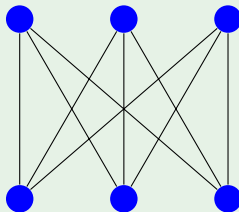
- These graphs generalise the complete graphs $K_n = \underbrace{K_{1, \dots, 1}}_n$

Example

K_5 :



$K_{3,3}$:



Exercise

6.2.14 Which complete graphs K_n have an Euler circuit?
When do bipartite, 3-partite complete graphs have an Euler circuit?

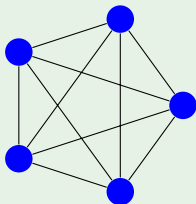
K_n has an Euler circuit for n odd

$K_{m,n}$ — when both m and n are even

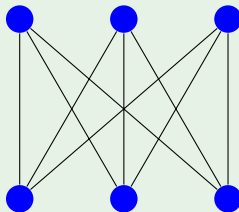
$K_{p,q,r}$ — when $p+q, p+r, q+r$ are all even, ie. when p, q, r are all even or all odd

Example

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Exercise

6.2.14 Which complete graphs K_n have an Euler circuit?
When do bipartite, 3-partite complete graphs have an Euler circuit?

K_n has an Euler circuit for n odd

$K_{m,n}$ — when both m and n are even

$K_{p,q,r}$ — when $p+q, p+r, q+r$ are all even, ie. when p, q, r are all even or all odd

Vertex Traversal

Definition

- **Hamiltonian path** visits every vertex of graph exactly once
- **Hamiltonian circuit** visits every vertex exactly once except the last one, which duplicates the first

(Named after Sir William Rowan Hamilton (Ireland), 1805–1865)

NB

Finding such a circuit, or proving it does not exist, is a *difficult* problem — the worst case is NP-complete.

Examples (when the circuit exists)

- n -cube; Hamiltonian circuit = *Gray code*
- complete graphs K_m ; bipartite graphs $K_{m,n}$ if $m = n$
- Knight's tour on a chessboard (incl. rectangular boards)

Exercise

6.5.5(a) How many Hamiltonian circuits does $K_{n,n}$ have?

Let $V = V_1 \dot{\cup} V_2$

- start at any vertex in V_1
- go to any vertex in V_2
- go to any *new* vertex in V_1
-

There are $n!$ ways to order each part and two ways to choose the 'first' part, implying $c = 2(n!)^2$ circuits.

Exercise

6.5.5(a) How many Hamiltonian circuits does $K_{n,n}$ have?

Let $V = V_1 \dot{\cup} V_2$

- start at any vertex in V_1
- go to any vertex in V_2
- go to any *new* vertex in V_1
-

There are $n!$ ways to order each part and two ways to choose the 'first' part, implying $c = 2(n!)^2$ circuits.

Given a graph it is nontrivial to verify that there is no Hamiltonian circuit: there is nothing obvious to specify that could assure us about this property.

In contrast, if a circuit is given, it is immediate to verify that it is a Hamiltonian circuit.

These situations demonstrate the often enormous discrepancy in difficulty of 'proving' versus (simply) 'checking'.

Colouring

Informally: assigning a “colour” to each vertex (e.g. a node in an electric or transportation network) so that the vertices connected by an edge have different colours.

Formally: A mapping $c : V \longrightarrow [1 .. n]$ such that for every $e = \{v, w\} \in E$

$$c(v) \neq c(w)$$

The minimum n sufficient to effect such a mapping is called the **chromatic number** of a graph $G = (E, V)$ and is denoted $\chi(G)$.

NB

This notion is extremely important in operations research, esp. in scheduling.

There is a dual notion of ‘edge colouring’ — two edges that share a vertex need to have different colours. Curiously enough, it is much less useful in practice.

Properties of the Chromatic Number

- $\chi(K_n) = n$
- If G has n vertices and $\chi(G) = n$ then $G = K_n$

Proof.

Suppose that G is 'missing' the edge $\{v, w\}$, as compared with K_n . Colour all vertices, except w , using $n - 1$ colours. Then assign to w the same colour as that of v . □

- If $\chi(G) = 1$ then G is totally disconnected: it has 0 edges.
- If $\chi(G) = 2$ then G is bipartite.
- For any tree $\chi(T) = 2$.
- For any cycle C_n its chromatic number depends on the parity of n — for n even $\chi(C_n) = 2$, while for n odd $\chi(C_n) = 3$.

Cliques

Graph (V', E') *subgraph* of (V, E) — $V' \subseteq V$ and $E' \subseteq E$.

Definition

A **clique** in G is a *complete* subgraph of G . A clique of k nodes is called *k-clique*.

The size of the largest clique is called the *clique number* of the graph and denoted $\kappa(G)$.

Theorem

$$\chi(G) \geq \kappa(G).$$

Proof.

Every vertex of a clique requires a different colour, hence there must be at least $\kappa(G)$ colours. □

However, this is the only restriction. For any given k there are graphs with $\kappa(G) = k$, while $\chi(G)$ can be arbitrarily large.

NB

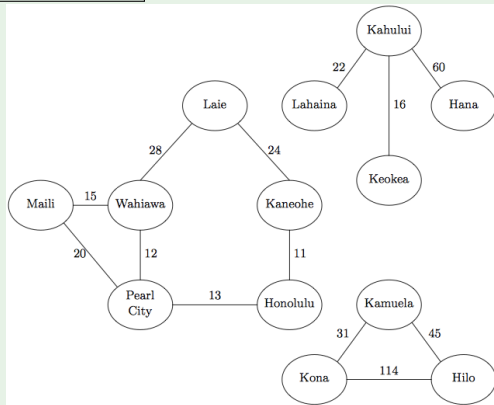
This fact (and such graphs) are important in the analysis of parallel computation algorithms.

- $\kappa(K_n) = n$, $\kappa(K_{m,n}) = 2$, $\kappa(K_{m_1, \dots, m_r}) = r$.
- If $\kappa(G) = 1$ then G is totally disconnected.
- For a tree $\kappa(T) = 2$.
- For a cycle C_n
 $\kappa(C_3) = 3$, $\kappa(C_4) = \kappa(C_5) = \dots = 2$

The difference between $\kappa(G)$ and $\chi(G)$ is apparent with just $\kappa(G) = 2$ — this does not imply that G is bipartite. For example, the cycle C_n for any odd n has $\chi(C_n) = 3$.

Exercise

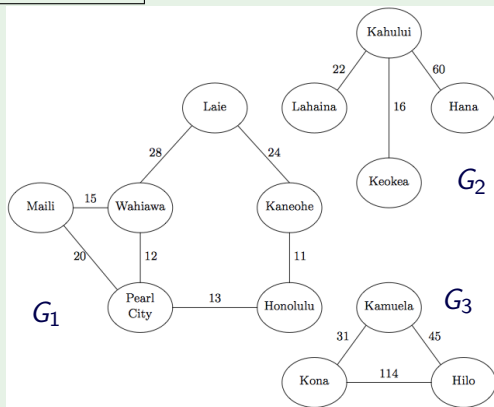
9.10.1 (Aho & Ullman)



$\chi(G)$? $\kappa(G)$? A largest clique?

Exercise

9.10.1 (Aho & Ullman)



$$\chi(G_1) = \kappa(G_1) = 3; \quad \chi(G_2) = \kappa(G_2) = 2; \quad \chi(G_3) = \kappa(G_3) = 3$$

Exercise

9.10.3 (Aho & Ullman) Let $G = (V, E)$ be an undirected graph. What inequalities must hold between

- the maximal $\deg(v)$ for $v \in V$
- $\chi(G)$
- $\kappa(G)$

$$\max_{v \in V} \deg(v) + 1 \geq \chi(G) \geq \kappa(G)$$

Exercise

9.10.3 (Aho & Ullman) Let $G = (V, E)$ be an undirected graph. What inequalities must hold between

- the maximal $\deg(v)$ for $v \in V$
- $\chi(G)$
- $\kappa(G)$

$$\max_{v \in V} \deg(v) + 1 \geq \chi(G) \geq \kappa(G)$$

Planar Graphs

Definition

A graph is **planar** if it can be embedded in a plane without its edges intersecting.

Theorem

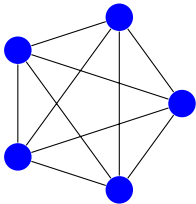
If the graph is planar it can be embedded in a plane (without self-intersections) so that all its edges are straight lines.

NB

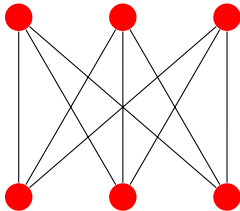
This notion and its related algorithms are extremely important to VLSI and visualising data.

Two minimal nonplanar graphs

K_5 :

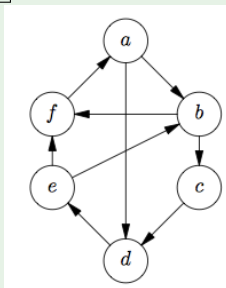


$K_{3,3}$:



Exercise

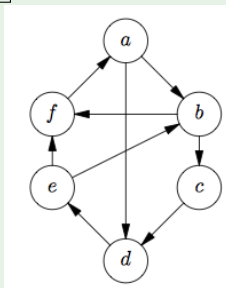
9.10.2 (Aho & Ullman)



Is (the undirected version of) this graph planar? Yes

Exercise

9.10.2 (Aho & Ullman)

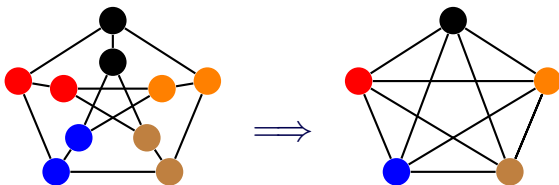


Is (the undirected version of) this graph planar? Yes

Theorem

If graph G contains, as a subgraph, a nonplanar graph, then G itself is nonplanar.

A *minor* of a graph is any graph obtained by repeatedly deleting vertices, deleting edges and merging *adjacent* vertices.



Theorem

A graph is nonplanar iff it contains K_5 or $K_{3,3}$ as a minor.

Theorem

K_n for $n \geq 5$ is nonplanar.

Proof.

It contains K_5 : choose any five vertices in K_n and consider the subgraph they define. □

Theorem

$K_{m,n}$ is nonplanar when $m \geq 3$ and $n \geq 3$.

Proof.

They contain $K_{3,3}$ — choose any three vertices in each of two vertex parts and consider the subgraph they define. □

Question

Are all $K_{m,1}$ planar?

Answer

Yes, they are trees of two levels — the root and m leaves.

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Yes, they are trees of two levels — the root and m leaves.

Question

Are all $K_{m,2}$ planar?

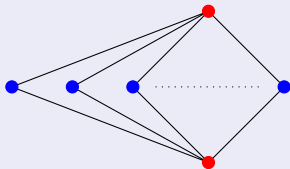
Question

Are all $K_{m,2}$ planar?

Answer

Yes; they can be represented by “glueing” together two such trees at the leaves.

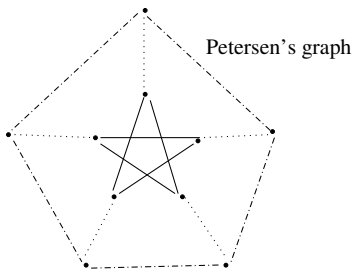
Sketching $K_{m,2}$:



Also, among the k -partite graphs, planar are $K_{2,2,2}$ and $K_{1,1,m}$. The latter can be depicted by drawing one extra edge in $K_{2,m}$, connecting the top and bottom vertices.

NB

Finding a 'basic' nonplanar obstruction is not always simple.



(Julius Petersen (Denmark), 1839–1910)

It contains both K_5 and $K_{3,3}$ as a minor (cf. Slide 39) while it does not directly contain either of them. (Try to find $K_{3,3}$ as a minor!)

Summary

- Graphs, trees, vertex degree, connected graphs, connected components, paths, cycles C_n
- Graph isomorphisms, automorphisms
- Special graphs: complete K_n , complete bi-, k -partite $K_{m_1, \dots, m_k}$
- Traversals
 - Euler paths and circuits (edge traversal)
 - Hamiltonian paths and circuits (vertex traversal)
- Graph properties:
chromatic number $\chi(G)$, clique number $\kappa(G)$, planarity

Coming up ...

- Mid-term test (covers weeks 1-5)
- Ch. 4, Sec. 4.2-4.6 (Induction and Recursion)